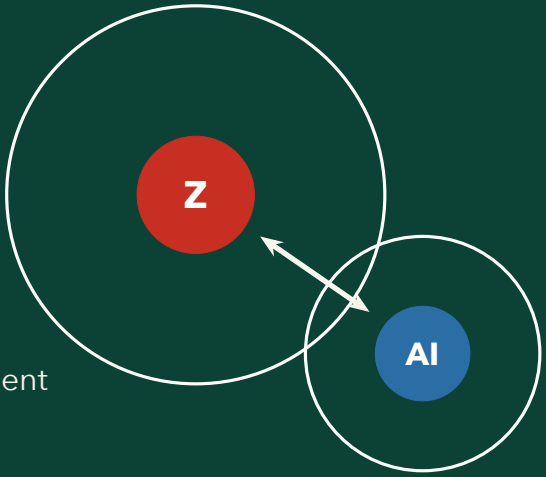
AI WORKSHOP | SESSION 03

AI × Zotero

Pick apps your AI can use:

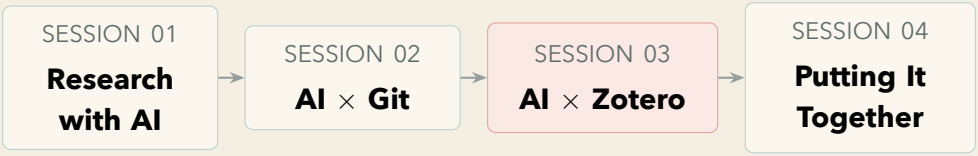
a literature library for you and your agent



Shih-Yang Lin | Department of Economics, National Dong Hwa University

October 14, 2026 | Room D410

Where We Are in the Series



Git tracks

how your research output changes

Zotero tracks

where your research claims come from

Workshop page: shihyanglin.github.io/ai-workshop

What You Will Build Today

Beaver beaverapp.ai

An AI assistant that lives inside Zotero.

- Cited answers from your library
- Highlight and annotate PDFs
- Organize tags and collections
- Batch work across many papers

Plus Max: $20/month

Today: Codex + Zotero + skills

The same pattern, built by you.

- `query-zotero-library`
- `annotate-zotero-pdf`
- `manage-zotero-library`
- Any agent, any app it can reach

No extra subscription

Leave with most of a $20/month tool, on the AI plan you already have.

Beaver, "Pricing," accessed October 14, 2026.

Today's Plan

01 | Why apps still matter
LLMs do not replace apps. They change *which* apps we pick.

02–04 | Set up, use, check
Install and set up sync while you listen. Then learn Zotero for humans and test your sync.

05 | Zotero + LLM
Let Codex query, highlight, and organize your own library.

06 | Wrap-up
One workflow you can repeat every day.

01

Why Apps Still Matter

LLMs do not replace apps. They change which apps we should pick.

Recall Session 01: What Makes AI Different

	Codifiable	Noncodifiable
Manual	Robots	Physical AI
Cognitive	**Enterprise software**	**AI**

Note: AI's defining feature is cognitive work that relies on noncodifiable knowledge. Codifiable cognitive work was already automated by software, and still is.

- **Codifiable** knowledge can be written down as exact rules.
  - Gives **deterministic** results.
  - **Costs less**: a specialized tool is a tiny model for one task, while a general LLM uses billions of parameters for the same job.
- **Noncodifiable** (tacit) knowledge cannot, and that is where AI is new.

Ide and Talamàs (2025, *JPE*), Table 1.

LLMs Do Not Make Apps Obsolete

Ask an LLM to *be* your reference manager

- Invented DOIs and authors
- Page numbers that drift
- Citation format changes every run
- Forgets your library next session

What an app guarantees

- Exact, structured records
- Same output every time
- Persistent storage on your disk
- Fast, cheap, no tokens spent

Codifiable work wants an App. Noncodifiable work wants an LLM.

Split Literature Work into Tasks

Codifiable → App

- Capture metadata
- Link the PDF to its record
- Format citations
- Sync across devices
- Store highlights, tags, and collections exactly

Noncodifiable → LLM + you

Find and understand

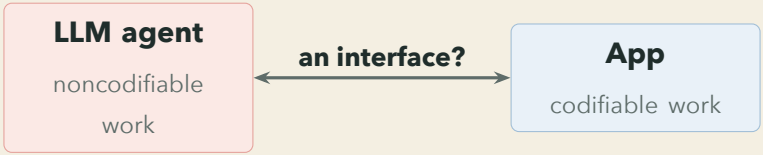
- Find a paper from a vague memory
- Explain a paper in plain words
- Answer questions with page citations
- Make reading guides and quizzes

Organize and connect

- Highlight by my own color rules
- Sort papers into collections
- Build a literature matrix
- Map how papers connect

Right panel: most NotebookLM features, applied to your own Zotero library.

Pick Apps Your Agent Can Reach



Three features of an AI-friendly app (from Session 01):

1. **Open data**: plain text or documented formats, not a black box.
2. **An API or CLI**: the agent can call it without clicking.
3. **Local and scriptable**: the agent can run it on your own machine.

An app your agent cannot reach makes you the copy-paste layer.

Example: ReadCube Papers vs. Zotero

	ReadCube Papers	Zotero
Reading and capture	**Excellent**	**Good**
Citation plugin	**Yes**	**Yes**
Where data lives	Vendor cloud	Your disk (SQLite + files)
Data format	Closed; export only	Open source, documented
Agent access	**No local API for agents**	**Local API on your laptop**

Human only

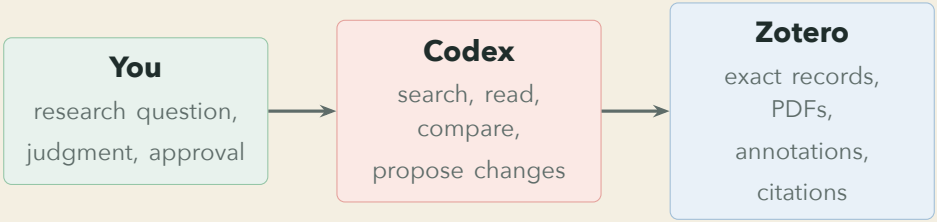
Both are fine.
Pick the one you enjoy.

Human + agent

Zotero, because the agent can read and write the library.

Based on the instructor's own use of both apps; check current vendor documentation.

The Division of Labor We Want



- Zotero **stores** what must be exact.
- Codex **orchestrates** what needs tacit judgment.
- You **decide** what counts as evidence and what gets written.

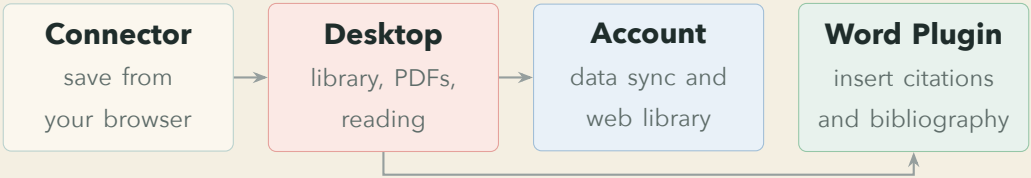
Same idea as Session 01: AI as the orchestrator, software as the SOP.

02

Install and Set Up

Start the downloads now. Set up sync while you listen to the next part.

Install Four Pieces of Zotero



1. Download Zotero Desktop at `zotero.org/download` (includes the Word plugin).
2. Install the Zotero Connector for your browser (same page).
3. Register a free account at `zotero.org/user/register`.

Zotero, "Quick Start Guide" and "Zotero Connectors."

Turn On the Local API

In Zotero Desktop

1. Open Settings.
2. Go to Advanced.
3. Check “Allow other applications on this computer to communicate with Zotero.”

Keep it local

- Zotero must be running.
- Only programs on this computer can connect, with no password.

Check it works (Terminal):

```
curl "http://localhost:23119/api/users/0/items?limit=1"
```

Any JSON reply (even `[]`) means the agent can now reach your library.

Zotero, “Local API.”

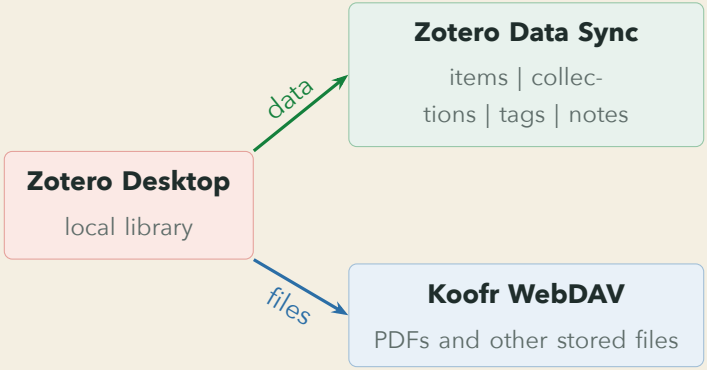
Install the Research Skills for Your Agent

```
git clone https://github.com/errard70101/social-science-research-skills.git
cd social-science-research-skills
python -m pip install -e .
python scripts/install.py --skill annotate-zotero-pdf --target codex
```

- One command installs three skills; the other two are dependencies.
- Open a **new** Codex session and check that these skills are listed: `query-zotero-library`, `manage-zotero-library`, `annotate-zotero-pdf`.
- Other agents: use `--target claude`, `copilot`, `opencode`, or `antigravity`.

github.com/errard70101/social-science-research-skills

Zotero Sync Has Two Pipelines



- Your Zotero account syncs metadata, with or without file sync.
- Koofr replaces only the file storage for your personal library.
- **Using Koofr does not mean you stop using Zotero's servers.**

Zotero, "Syncing," updated June 17, 2026.

Choose Your File Sync

	Zotero Storage	Koofr WebDAV
Setup	Easiest, official	Needs Koofr + WebDAV
Personal library	Supported	Supported
Group libraries	**Supported**	**Not supported**
PDFs in web library	Yes	No
Free space	300 MB	Depends on Koofr plan
Support	Zotero	Each vendor separately

Pick Zotero Storage for simplicity and groups; pick Koofr for more personal space.

Zotero, “Syncing”; Koofr, “Koofr with Zotero via WebDAV.”

Koofr Step 1: Create an App Password

1. Sign up at `koofr.eu`.
2. Turn on two-factor authentication.
3. Open Account Settings, then Preferences.
4. Under Password, create an app-specific password.
5. Name it `Zotero–WebDAV`.

Not your login password

WebDAV requires an app-specific password.

It may be shown only once. Save it in a password manager now.

Koofr Help, “How Do I Connect a Service to Koofr through WebDAV?”

Koofr Step 2: Enter WebDAV in Zotero

Path

Zotero Settings
↓
Sync
↓
File Syncing
↓
WebDAV

Connection

URL	`https://app.koofr.net/dav/Koofr`
Username	your Koofr email
Password	the app-specific password

Then click **Verify Server**.

Zotero stores files in a `/zotero` folder under that URL. If asked to create it, say yes.

Koofr Help; Zotero, “Sync Preferences.”

Install Checklist: Keep This Open

- ☐ Zotero Desktop opens.
- ☐ Connector icon appears in the browser.
- ☐ Logged in under Settings → Sync.
- ☐ Koofr `Verify Server` succeeds.
- ☐ Local API check returns JSON.
- ☐ Codex lists the three Zotero skills.

Pick one test paper now

A journal article page with a DOI and a PDF you can legally download.

Stuck? Raise your hand and keep listening. We will not need the agent until Part 05.

03

Zotero for Humans

Collect, check, organize, read, cite. The codifiable half of the work.

A Downloads Folder Is Not a Library

What most hard drives look like

- `paper.pdf`
- `paper-final.pdf`
- `download(7).pdf`
- Authors, years, DOIs scattered

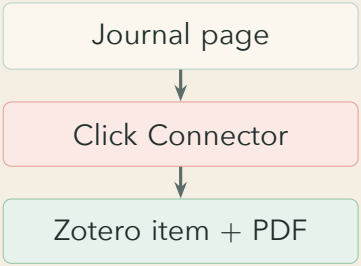
What research actually needs

- Correct, structured metadata
- PDFs, highlights, notes linked
- Search, cite, sync, verify
- An agent can find the page

Zotero links every PDF to its metadata, notes, and citations.

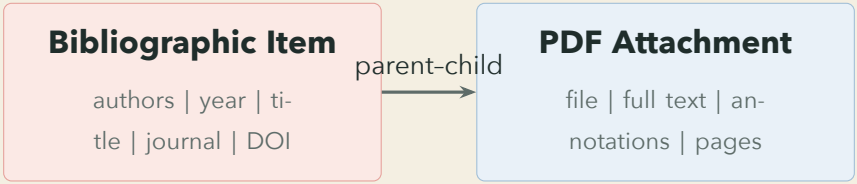
Save Your First Paper with the Connector

1. Keep Zotero Desktop open.
2. Open the article's journal page.
3. Click the Connector icon next to the address bar.
4. Save to `My Library`.
5. In Desktop, check the item and its PDF.



If the Connector saves only a "Web Page," you are not on the article page. Do not accept the wrong metadata.

One Paper, Two Objects



- Item without PDF: you can cite it, but the agent cannot read it.
- Standalone PDF: full text, but no reliable identity.
- **Goal:** the correct item linked to the correct PDF.

Check Metadata Before You Cite

Minimum checklist

- Item type
- Title
- Authors or editors
- Year, journal, volume, issue
- DOI or stable URL

Why check now?

- Typos spread to every citation
- Bad capitalization pollutes BibTeX
- The wrong PDF misleads the agent
- Duplicates split notes and tags

Collections and Tags Answer Different Questions

Collections | Which project is it for?

- `0 Working Projects`
- `Inflation Inequality`
- `Teaching / Macrofinance`

One item can live in several collections.

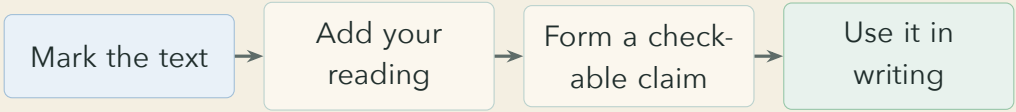
Tags | What is its status or theme?

- `to-read`
- `reviewed`
- `identification`

Tags work across projects.

Start small: a few collections first; add stable tags once a need repeats.

Reading Is More Than Highlighting



- An annotation keeps the **exact position and page**.
- A note keeps **your interpretation, questions, and uses**.
- An AI summary must never pose as the author's own words.

Let Zotero Maintain Your Bibliography

Word, LibreOffice, Google Docs

1. Add/Edit Citation
2. Search by author or title
3. Choose a citation style
4. Add/Edit Bibliography

LaTeX

1. Export BibTeX or BibLaTeX
2. Keep citation keys stable
3. Track the `.bib` file in Git
4. Later: try Better BibTeX

Do not hand-edit a bibliography as plain text. Fix the Zotero item, then refresh.

Checkpoint 1: One Usable Piece of Evidence

1. Save one paper from its article page with the Connector.
2. Confirm the item and its PDF attachment.
3. Check authors, year, journal, and DOI.
4. Create a `Workshop Demo` collection and add the item.
5. Add one highlight or note.

Done when: you can open the PDF from the item, and find the item from the PDF.

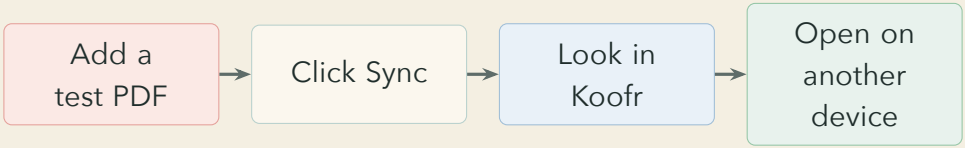
If the Connector or PDF download fails, fix the item before moving on.

04

Check Your Sync

Prove your PDFs sync, and keep a backup that sync cannot replace.

After Verify Server, Run a Visible Test



- You may see `.zip` files in Koofr's `zotero` folder. That is normal.
- Never rename, move, or tidy those files by hand.
- Real success: another Zotero install downloads and opens the PDF.

Koofr, "Koofr with Zotero via WebDAV," updated April 2025.

WebDAV Covers Your Personal Library Only

My Library

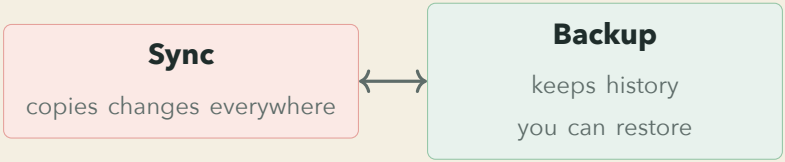
- Metadata: Zotero Data Sync
- PDFs: Koofr WebDAV
- Good for personal research

Group Libraries

- WebDAV does not sync group files
- Group files need Zotero Storage
- Access set by group settings

Zotero, “Syncing,” section “WebDAV.”

Sync Is Not Backup



Deletions, mistakes, and corruption sync too. Sync cannot replace a backup.

Never put your live Zotero data directory inside a Dropbox, iCloud, Google Drive, or Koofr sync folder.

To back up the data directory, quit Zotero first, then copy it somewhere independent.

Zotero, "Syncing," section "Alternative Syncing Solutions."

Checkpoint 2: Prove Sync Actually Works

1. Logged in to your Zotero account; data sync shows no errors.
2. File sync for My Library is set to WebDAV.
3. Koofr URL, email, and app password are filled in.
4. `Verify Server` succeeds.
5. The test PDF syncs and opens on another device or a clean install.

Success: the PDF opens inside Zotero.

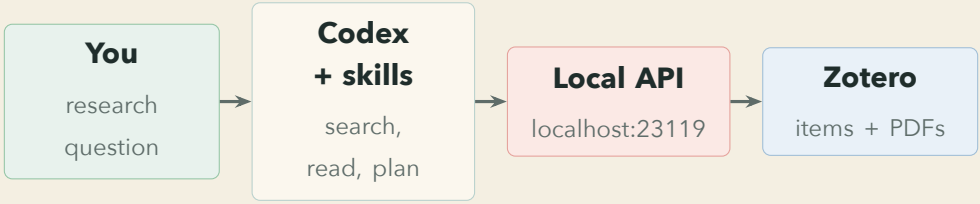
Not yet: you only see a file name in Koofr.

05

Zotero + LLM

Now let the agent reach the library. It reads freely; it writes only with your approval.

How Codex Reaches Your Library



- No cloud upload of your library; everything stays on your laptop.
- No Web API key needed for reading.
- A **skill** is a written SOP: it tells the agent exactly how to call the API safely.

Zotero, "Local API," updated July 29, 2026.

Three Skills, Mapped to the Task Split

Skill	LLM does	Zotero does	Boundary
`query-zotero-library`	find, read, compare	return items, PDFs, notes	read-only
`annotate-zotero-pdf`	locate the passage	store the highlight exactly	no edits or deletes
`manage-zotero-library`	propose tags and collections	apply the approved plan	no metadata, no PDFs

Each skill pairs one **noncodifiable** judgment with one **codifiable** record.

Query Your Library, Not the Chat Window

Approach	Convenient because	Research risk
Paste PDFs into chat	Starts immediately	Short-lived context, scattered sources
Let AI search the web	Broad coverage	Not the version you trust or read
Search the repo	Close to your output	PDFs usually live elsewhere
Query Zotero	**Your library is the entry point**	Still check PDF pages

Demo 1: A Source-Grounded Literature Query

```
Use $query-zotero-library.

Which papers in my library criticize business cycle accounting,
and what are their main arguments?

Compare at most three, using only the PDF text. List any gaps
in the evidence.
```

- Ask the way you remember it: a vague question is enough.
- Name the **evidence boundary**: PDF text only.
- Every claim comes back with a **link to its page**; the skill adds it.

One Step Back to the Source

```
zotero://open-pdf/library/items/<key>?page=<page>&annotation=<highlight>
```

Read it now

"Open the evidence for claim 2."

The agent runs `open` (macOS), `start` (Windows), or `xdg-open` (Linux), and Zotero jumps to the page.

Works in any agent that can run a command.

Read it later

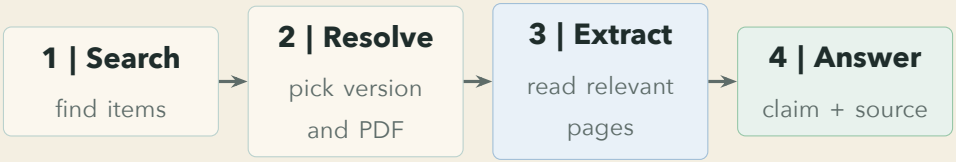
"Save this comparison as a note, with the links."

Click a link in Obsidian or any Markdown note, and Zotero jumps to the page.

Keys sync, so links work on every device.

Chat windows may strip `zotero://` links; let the agent open them instead.

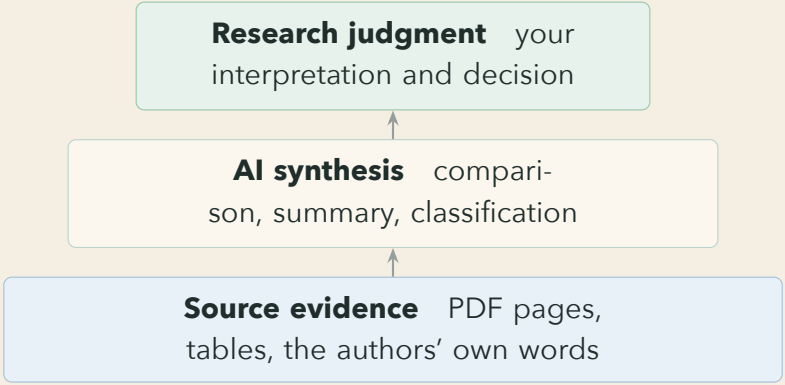
What Happens Behind the Query



Metadata only proves a paper exists. Claims about content must come from the PDF pages.

No PDF, failed text extraction, or thin evidence? The right answer states the limit; it does not invent a plausible summary.

Read the Answer in Three Layers



- An item key does not mean the full text was read.
- A PDF summary does not mean the claim is proven.
- An AI draft is not a research conclusion.

Demo 2: Let Codex Highlight the Evidence

```
Use $annotate-zotero-pdf.

Read my highlight color rules in my Obsidian vault (06 SOPs).
Then highlight the key claims and evidence in Chari, Kehoe, and
McGrattan (2007), following those rules.

Add a short margin comment and the tag workshop-demo to each.
Show the full batch plan and ask for one exact plan ID. Do not
write before I approve.
```

- **LLM**: decides whether a sentence is a claim, evidence, or a problem, and picks the color.
- **App**: stores each mark at exact coordinates you can reopen anytime.
- Your rules live in a note, so the agent applies them the same way every time.

One Batch, One Approval



- **No per-annotation clicks:** one reviewed plan ID covers the batch.
- If the library or PDF changes, the old plan expires; plan again.
- Same passage twice on a page? Add context; never guess.
- Afterward, check both the Reader view and the annotation sidebar.

Zotero, "Local API" and "Write Requests"; `annotate-zotero-pdf` safety contract.

Demo 3: Organize with a Reviewable Plan

```
Use $manage-zotero-library.

Find the papers from Demo 1 and propose a plan:
  1. add them to the Workshop Demo collection;
  2. add the tag workshop-reviewed.

Show before and after for each paper and the exact plan ID.
Do not write until I approve.
```

- Collection membership and tags are easy to undo.
- Title, authors, DOI, and PDFs stay untouched.
- Practice on a test collection before a real project.

Before You Approve, Read Four Columns

Item	Before	After	What you judge
Paper A	no demo tag	add demo tag	Right version?
Paper B	not in collection	add to collection	Right collection?
Paper C	already reviewed	unchanged	Really a no-op?

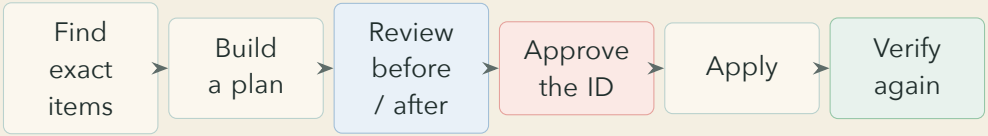
Plan ID

Binds the exact changes you saw.

Library version

If the library changes, the plan expires.

Six Gates for Every Write



Codex writes only after you approve the **exact plan ID**.

Timeout or server error mid-write? Re-read the library first. A blind retry can apply the same change twice.

Zotero’s Authorization Dialog Has Three Choices

Allow

Permits only the next successful write.

Use this in class.

Always Allow

Remembers an unscoped local key.

Store it in the OS keychain.

Deny

Refuses this write.

Always available.

Codex should never pick Always Allow for you, or put a key in a prompt, plan, log, or repo.

Zotero, “Local API,” section “Authorizing Writes.”

Beyond Zotero: More Skills in the Same Repo

Skill	What it does
`manage-latex-bibliography`	Finds missing citations in a LaTeX project and adds verified BibTeX entries
`summarize-academic-paper`	Writes a two-page LaTeX summary of an economics paper for seminar prep
`literature-search-repec`	Searches IDEAS/RePEc for working papers, journal articles, and citations
`rename-and-organize-references`	Proposes author-year-title file names for a messy PDF folder

Same pattern every time: the agent proposes, a deterministic tool executes, you review.

github.com/errard70101/social-science-research-skills

Checkpoint 3: One Query, One Approved Write

1. Pick a small, clear research question.
2. Ask Codex to search Zotero first, not to answer from memory.
3. Limit it to 2–3 highly relevant papers.
4. Check that every item key and page matches the PDF.
5. Approve one tag or annotation plan, choose `Allow`, and verify in Zotero.

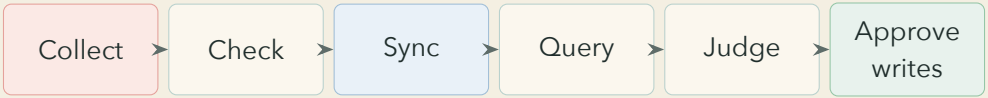
Done when: you can say who proposed, what you approved, and what actually changed.

06

Putting It Together

Choose tools by task. Keep every claim traceable to its source.

One Complete Zotero × Codex Workflow



Zotero keeps (codifiable)

- Bibliographic identity
- PDFs and page numbers
- Annotations and notes

Codex helps (noncodifiable)

- Search and compare
- Extract evidence
- Propose reviewable changes

Three Takeaways

1. **Split work by knowledge type.** Codifiable work goes to apps; noncodifiable work goes to LLMs.

   Zotero stores the citation; Codex judges which papers matter.

2. **Pick apps your agent can reach.** Open data, an API, and local access let the two work together.

   Zotero has a Local API; ReadCube does not.

3. **Stay the judge.** The agent searches, explains, and proposes; you verify and approve.

   Every claim traces to a page; every write needs your plan approval.

Next session: combine Git, Zotero, skills, and agents into one repeatable workflow.

Homework: Redo It with Your Own Research

1. Create a collection for a real research project.
2. Add 5–10 core papers with checked metadata.
3. Sync files with Koofr and make one independent backup.
4. Ask Codex to compare three papers' identification strategies.
5. Review and approve one tag, collection, or annotation plan.

References i

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